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Unit 02 Study Notes

MAP2302 Differential Equations

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Topics covered: Higher-Order Linear ODEs · Reduction of Order · Constant-Coefficient Equations · Undetermined Coefficients (Superposition & Annihilator)

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Appendix: Quick Reference

A.1 Integration by Parts Shortcuts

Recall the IBP formula: $\int u\,dv = uv - \int v\,du$. For repeated application use the **tabular (tic-tac-toe) method**: alternate signs $+, -, +, -, \ldots$, differentiate the left column to zero, integrate the right column, then multiply diagonally.

Integral	Result	Notes
$\int xe^{ax}\,dx$	$\frac{e^{ax}}{a}\left(x - \frac{1}{a}\right) + C$	$u = x,\ dv = e^{ax}dx$
$\int x^2e^{ax}\,dx$	$\frac{e^{ax}}{a}\left(x^2 - \frac{2x}{a} + \frac{2}{a^2}\right) + C$	tabular method
$\int xe^{ax}\cos(bx)\,dx$	use tabular twice	cycles after 2 steps
$\int e^{ax}\cos(bx)\,dx$	$\frac{e^{ax}(a\cos bx + b\sin bx)}{a^2+b^2} + C$	IBP twice, solve for I
$\int e^{ax}\sin(bx)\,dx$	$\frac{e^{ax}(a\sin bx - b\cos bx)}{a^2+b^2} + C$	IBP twice, solve for I
$\int x\cos(bx)\,dx$	$\frac{\cos(bx)}{b^2} + \frac{x\sin(bx)}{b} + C$	$u = x,\ dv = \cos(bx)dx$
$\int x\sin(bx)\,dx$	$\frac{\sin(bx)}{b^2} - \frac{x\cos(bx)}{b} + C$	$u = x,\ dv = \sin(bx)dx$
$\int x^ne^{ax}\,dx$	$\frac{x^ne^{ax}}{a} - \frac{n}{a}\int x^{n-1}e^{ax}\,dx$	reduction formula

A.2 Trigonometric Identities

Reciprocal and Quotient Identities

$$\csc x = \frac{1}{\sin x}, \qquad \sec x = \frac{1}{\cos x}, \qquad \cot x = \frac{1}{\tan x} = \frac{\cos x}{\sin x}, \qquad \tan x = \frac{\sin x}{\cos x}.$$

Pythagorean Identities

$$\sin^2 x + \cos^2 x = 1, \qquad 1 + \tan^2 x = \sec^2 x, \qquad 1 + \cot^2 x = \csc^2 x.$$

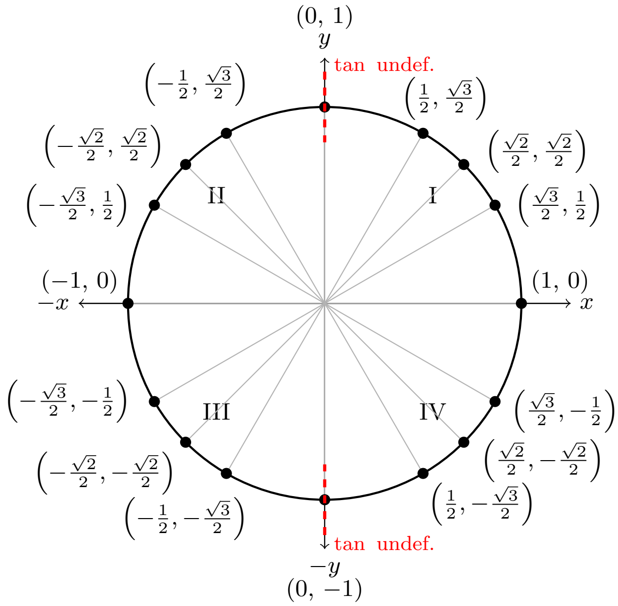
Double-Angle Identities

$$\sin(2x) = 2\sin x\cos x, \qquad \cos(2x) = \cos^2 x - \sin^2 x = 1 - 2\sin^2 x = 2\cos^2 x - 1.$$

Power-Reduction Identities

$$\sin^2 x = \frac{1 - \cos(2x)}{2}, \qquad \cos^2 x = \frac{1 + \cos(2x)}{2}.$$

A.3 Unit Circle and Domain Restrictions



Why $\tan x$ causes problems at $x = \pm\pi/2$. Since $\tan x = \sin x/\cos x$, the function is undefined wherever $\cos x = 0$, namely at $x = \pm\frac{\pi}{2}, \pm\frac{3\pi}{2}, \ldots$ On the unit circle these correspond to the points $(0, 1)$ and $(0, -1)$ where the x-coordinate (i.e. $\cos x$) is zero.

For an ODE such as $y'' + (\tan x)\, y = e^x$, the coefficient $\tan x$ is **continuous on any open interval not containing these points**. The largest such interval centered at $x = 0$ is therefore $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, which is exactly the interval guaranteed by the existence and uniqueness theorem.

Function	**Undefined at**	**Continuous on**
$\tan x = \sin x/\cos x$	$x = \frac{\pi}{2} + n\pi$	each $\left(-\frac{\pi}{2} + n\pi,\ \frac{\pi}{2} + n\pi\right)$
$\cot x = \cos x/\sin x$	$x = n\pi$	each $(n\pi,\ (n+1)\pi)$
$\sec x = 1/\cos x$	$x = \frac{\pi}{2} + n\pi$	same as $\tan x$
$\csc x = 1/\sin x$	$x = n\pi$	same as $\cot x$

$n \in \mathbb{Z}$

1 Higher Order Linear Differential Equations

1.1 Introduction to Higher Order Linear ODEs

1.1.1 General Form and Terminology

An n^{th}-order linear ODE has the form

$$a_0(x)\, y^{(n)} + a_1(x)\, y^{(n-1)} + \cdots + a_n(x)\, y = g(x),$$

where the coefficient functions $a_k(x)$ and the forcing term $g(x)$ are given on an interval and $a_0(x) \neq 0$. If $g(x) = 0$ the equation is **homogeneous**; otherwise it is **non-homogeneous**.

1.1.2 Existence and Uniqueness of Solutions

For a linear n^{th}-order ODE with continuous coefficients $a_k(x)$ and forcing term $g(x)$ on an interval I, and prescribed initial conditions $y(x_0)$, $y'(x_0)$, $\ldots$, $y^{(n-1)}(x_0)$, there exists a *unique* solution on I.

This is the higher-order version of the Picard–Lindelöf existence and uniqueness theorem, and guarantees well-posed initial value problems whenever the coefficients are continuous.

1.1.3 The Superposition Principle

If $L[y] = 0$ is a linear homogeneous ODE and $y_1, \ldots, y_k$ are solutions, then any linear combination $c_1 y_1 + \cdots + c_k y_k$ is also a solution.

This turns the solution set into a **vector subspace**, which is why we seek a basis — a *fundamental set of solutions* — for it.

1.2 The Structure of Solutions to Homogeneous Equations

1.2.1 Linear Independence and Solution Sets

Solutions $y_1, \ldots, y_n$ are **linearly independent** on an interval if the only constants $c_1, \ldots, c_n$ satisfying $c_1 y_1 + \cdots + c_n y_n \equiv 0$ are $c_1 = \cdots = c_n = 0$.

For an n^{th}-order homogeneous linear ODE, any set of n linearly independent solutions **spans the entire solution space**.

1.2.2 Fundamental Sets of Solutions

A **fundamental set of solutions** is a set $\{y_1, \ldots, y_n\}$ of n linearly independent solutions. Every solution can then be written as

$$y(x) = c_1 y_1(x) + \cdots + c_n y_n(x),$$

where the constants c_i are determined by the initial conditions.

1.3 The Wronskian and Linear Independence

1.3.1 Definition and Computation of the Wronskian

For functions $y_1, \ldots, y_n$, the **Wronskian** is

$$W(y_1, \ldots, y_n)(x) = \det \begin{pmatrix} y_1 & y_2 & \cdots & y_n \\ y_1' & y_2' & \cdots & y_n' \\ \vdots & \vdots & \ddots & \vdots \\ y_1^{(n-1)} & y_2^{(n-1)} & \cdots & y_n^{(n-1)} \end{pmatrix}.$$

Build the matrix whose k^{th} row contains the $(k-1)^{\text{th}}$ derivatives of each function, then take the determinant.

1.3.2 Abel's Theorem

For $y'' + p(x)y' + q(x)y = 0$ with continuous p, q on I, Abel's formula gives

$$W(x) = W(x_0) \exp\left(- \int_{x_0}^{x} p(s)\, ds \right).$$

Consequently the Wronskian is either *never zero* or *identically zero* on I — linear independence is an all-or-nothing property on the interval.

1.3.3 Using the Wronskian to Verify Independence

If $W(y_1, \ldots, y_n)(x_0) \neq 0$ at some point x_0, then $y_1, \ldots, y_n$ are linearly independent on the interval. Conversely, if the functions are linearly dependent their Wronskian is identically zero.

2 Reduction of Order

2.1 Introduction to Reduction of Order

2.1.1 Motivation: When One Solution Is Known

Reduction of order is a technique for finding a second, linearly independent solution of $y''+P(x)y'+Q(x)y = 0$ when one nontrivial solution $y_1(x)$ is already known.

Rather than solving from scratch, the method reduces the problem to a **first-order** equation for an auxiliary function.

2.1.2 Deriving the Reduced Equation via Substitution

Given y_1, set $y(x) = v(x)\, y_1(x)$, substitute into the ODE, and simplify using the fact that y_1 already satisfies the equation. This yields a first-order ODE for $v'(x)$. Integrating gives the reduction formula

$$y_2 = y_1(x) \int \frac{e^{-\int P(x)\, dx}}{y_1^2(x)}\, dx.$$

2.2 Homogeneous Equations

2.2.1 Worked Example: Finding a Second Independent Solution

Template. If $y_1(x)$ solves $y'' + P(x)y' + Q(x)y = 0$, write $y = v\, y_1$, compute y' and y'', substitute, and reduce to a first-order linear ODE in v'. Solving gives $y_2 = v\, y_1$ such that $\{y_1, y_2\}$ is a fundamental set.

2.2.2 Verification via the Wronskian

After finding y_2, compute $W(y_1, y_2)(x)$. If $W(x_0) \neq 0$ at any point, then y_1 and y_2 are linearly independent and the general solution is $y = c_1 y_1 + c_2 y_2$.

2.3 Non-Homogeneous Equations

2.3.1 Adapting Reduction of Order for Non-Homogeneous Cases

For $y'' + P(x)y' + Q(x)y = F(x)$ with known homogeneous solution y_1, the substitution $y = v\, y_1$ converts the equation into a first-order ODE for v incorporating the forcing term $F(x)$.

2.3.2 Worked Example: Second-Order Non-Homogeneous Equation

Template. Verify y_1 solves the homogeneous equation, set $y = v\, y_1$, derive and solve the ODE for v, obtain the particular solution $y_p = v\, y_1$, then write $y = y_c + y_p$.

3 Homogeneous Linear Equations with Constant Coefficients

3.1 Introduction and the Characteristic Equation

3.1.1 Deriving the Characteristic (Auxiliary) Equation

For $a_n y^{(n)} + \cdots + a_1 y' + a_0 y = 0$, assume $y = e^{rx}$, substitute, and cancel e^{rx} to obtain the **characteristic polynomial**

$$a_n r^n + a_{n-1} r^{n-1} + \cdots + a_1 r + a_0 = 0.$$

Its roots determine the basic exponential and oscillatory building blocks of the general solution.

3.1.2 General Solution Framework

Roots $r_1, \ldots, r_k$ (with multiplicities) produce solution terms of the form $x^m e^{r_j x}$ (real roots) or $x^m e^{\alpha x} \cos(\beta x)$ and $x^m e^{\alpha x} \sin(\beta x)$ (complex roots $\alpha \pm i\beta$).

Three cases: **distinct real roots**, **repeated real roots**, **complex conjugate roots**.

3.2 Case I: Distinct Real Roots

3.2.1 Solution Form and General Solution

If the characteristic equation has n distinct real roots $r_1, \ldots, r_n$, the general solution is

$$y(x) = c_1 e^{r_1 x} + c_2 e^{r_2 x} + \cdots + c_n e^{r_n x}.$$

3.2.2 Worked Examples

Template: $y'' - 3y' + 2y = 0 \;\Rightarrow\; r^2 - 3r + 2 = 0 \;\Rightarrow\; r = 1,\, 2 \;\Rightarrow\; y(x) = c_1 e^{x} + c_2 e^{2x}.$

3.3 Case II: Repeated Real Roots

3.3.1 Why Repetition Requires a Modified Solution

If a real root r has multiplicity m, the m linearly independent solutions are e^{rx}, xe^{rx}, $\ldots$, $x^{m-1}e^{rx}$. The factors of x compensate for the lack of enough distinct exponentials.

3.3.2 Reduction of Order Justification

Applying reduction of order to $y_1 = e^{rx}$ when r is a repeated root reveals that the second independent solution must be xe^{rx}. Repeating the argument explains $x^k e^{rx}$ for higher multiplicities.

3.3.3 Worked Examples

Template: $y'' - 4y' + 4y = 0 \;\Rightarrow\; (r-2)^2 = 0 \;\Rightarrow\; y(x) = (c_1 + c_2 x)\, e^{2x}.$

3.4 Case III: Complex Conjugate Roots

3.4.1 Euler's Formula and Real-Valued Solutions

For roots $\alpha \pm i\beta$ $(\beta \neq 0)$, Euler's formula $e^{i\beta x} = \cos(\beta x) + i\sin(\beta x)$ converts complex exponentials into a real solution pair: $e^{\alpha x}\cos(\beta x)$ and $e^{\alpha x}\sin(\beta x)$.

3.4.2 Extracting Sine and Cosine Forms

Taking real and imaginary parts of $e^{(\alpha+i\beta)x} = e^{\alpha x}(\cos\beta x + i\sin\beta x)$ gives the general real solution

$$y(x) = e^{\alpha x}\left(c_1\cos(\beta x) + c_2\sin(\beta x)\right).$$

3.4.3 Worked Examples

Template: $y'' + y = 0 \;\Rightarrow\; r^2 + 1 = 0 \;\Rightarrow\; r = \pm i \;\Rightarrow\; y(x) = c_1\cos x + c_2\sin x.$

4 Undetermined Coefficients — Superposition Approach

4.1 Non-Homogeneous Equations and the Structure of Solutions

4.1.1 The Complementary and Particular Solution

For $L[y] = g(x)$, the general solution is $y(x) = y_c(x) + y_p(x)$, where y_c solves $L[y] = 0$ (the **complementary solution**) and y_p is any particular solution of $L[y] = g(x)$.

4.1.2 Superposition Principle for Non-Homogeneous Equations

If $g(x) = g_1(x) + \cdots + g_k(x)$ and each $y_{p,i}$ solves $L[y] = g_i(x)$, then $y_p = y_{p,1} + \cdots + y_{p,k}$ solves $L[y] = g(x)$. Handle each forcing term separately and sum the results.

4.2 Selecting the Form of the Particular Solution

4.2.1 Trial Functions for Polynomials, Exponentials, and Sinusoids

Choose a trial y_p whose form mirrors $g(x)$: a polynomial of the same degree for polynomial forcing, Ae^{ax} for $g(x) = e^{ax}$, and $A\cos(bx) + B\sin(bx)$ for sinusoidal forcing. Substitute and equate coefficients to find the unknowns.

4.2.2 Table of Standard Trial Function Forms

Forcing $g(x)$	Trial form for y_p
$P_n(x)$	degree-n polynomial
$e^{ax}P_n(x)$	e^{ax} $\times$ degree-n polynomial
$\cos(bx)$ or $\sin(bx)$	$A\cos(bx) + B\sin(bx)$
$e^{ax}\cos(bx)$ or $e^{ax}\sin(bx)$	$e^{ax}(A\cos(bx) + B\sin(bx))$

4.3 Worked Examples

4.3.1 Polynomial Forcing Function

For $y'' - y = 3x^2$, try $y_p = Ax^2 + Bx + C$, compute $y_p'' - y_p$, and solve for A, B, C by matching coefficients with $3x^2$.

4.3.2 Exponential Forcing Function

For $y'' - 3y' + 2y = e^{2x}$, try $y_p = Ae^{2x}$ provided $r = 2$ is not a root of the characteristic equation; substitute and solve for A.

4.3.3 Sinusoidal Forcing Function

For $y'' + y = \cos x$, try $y_p = A\cos x + B\sin x$, substitute, and solve for A and B by matching $\cos x$ and $\sin x$ terms.

4.4 Modification Rule: Multiplying by x for Linear Independence

4.4.1 When the Trial Function Solves the Homogeneous Equation

If the initial trial $y_p^{(0)}$ is already a solution of the homogeneous equation, multiply it by x^k (where k is the multiplicity of the corresponding root) to obtain a trial linearly independent of all homogeneous solutions. This ensures y_p is not absorbed into y_c.

4.4.2 Worked Examples Applying the Modification Rule

For $y'' - 4y' + 4y = e^{2x}$: root $r = 2$ has multiplicity 2, so use $y_p = Ax^2e^{2x}$.

General rule: if $r = a$ has multiplicity k in the characteristic polynomial and $g(x) \sim e^{ax}$, the correct trial is $y_p = Ax^ke^{ax}$.

5 Undetermined Coefficients — Annihilator Approach

5.1 Differential Operators and Operator Notation

5.1.1 The D-Operator and Polynomial Operators

Define $D = d/dx$, so $Dy = y'$, $D^2 y = y''$, etc. A constant-coefficient ODE can be written as $P(D)\,y = g(x)$ for a polynomial P. This notation allows differential equations to be manipulated using polynomial algebra.

5.1.2 Factoring Differential Operators

If $P(D) = (D - r_1)(D - r_2)\cdots(D - r_n)$, then $P(D)\,y = 0$ decomposes into independent first-order equations $(D - r_k)\,y = 0$, giving exponential solutions $e^{r_k x}$. Factoring $P(D)$ parallels factoring the characteristic polynomial.

5.2 Annihilators

5.2.1 Definition: What Is an Annihilator?

A linear differential operator $A(D)$ is called an **annihilator** of $f(x)$ if $A(D)\,f(x) = 0$. For example, $D - a$ annihilates e^{ax}, and $D^2 + b^2$ annihilates $\sin(bx)$ and $\cos(bx)$.

5.2.2 Annihilators for Polynomials, Exponentials, and Sinusoids

- $P_n(x)$ (degree n) is annihilated by D^{n+1}.
- e^{ax} is annihilated by $D - a$.
- $\cos(bx)$ and $\sin(bx)$ are annihilated by $D^2 + b^2$.
- $e^{ax}\cos(bx)$ and $e^{ax}\sin(bx)$ are annihilated by $(D - a)^2 + b^2$.
- $e^{ax}P_n(x)$ is annihilated by $(D - a)^{n+1}$.

5.2.3 Finding the Annihilator of a Given Forcing Function

Decompose $g(x)$ into its basic types and combine the corresponding basic annihilators multiplicatively. For example, $x^k e^{ax}\cos(bx)$ is annihilated by $\left[(D - a)^2 + b^2\right]^{k+1}$.

5.3 Solving Non-Homogeneous Equations via Annihilators

5.3.1 Applying the Annihilator to Both Sides

For $P(D)\,y = g(x)$, choose $A(D)$ with $A(D)\,g(x) = 0$ and apply it to both sides:

$$A(D)\,P(D)\,y = 0.$$

5.3.2 Solving the Resulting Higher-Order Homogeneous Equation

Solve $A(D)\,P(D)\,y = 0$ by finding all characteristic roots from both P and A, then build the general solution from the associated exponential and sinusoidal terms.

5.3.3 Eliminating Complementary Terms to Isolate the Particular Solution

The portion of the combined general solution coming from roots of $P(D)$ is y_c; the remaining terms form the trial y_p. Substitute y_p back into $P(D)\,y = g(x)$ to determine its unknown coefficients. The final general solution is $y = y_c + y_p$.

5.4 Worked Examples

5.4.1 Example with Polynomial Forcing

For $y'' - y = x$: $g(x) = x$ is annihilated by D^2. Apply D^2 to get $D^2(D^2-1)\,y = 0$, roots $r = \pm 1,\, 0\,(m = 2)$. New roots give trial $y_p = Ax + B$; substitute to find A, B.

5.4.2 Example with Exponential Forcing

For $y'' - 3y' + 2y = e^x$: $D - 1$ annihilates e^x. Apply $D - 1$ to get $(D-1)(D^2 - 3D + 2)\,y = 0$, roots $r = 1\,(m = 2)$, $r = 2$. Extra $r = 1$ factor gives trial $y_p = Axe^x$; substitute to find A.

5.4.3 Example Requiring a Higher-Order Annihilator

For $g(x) = xe^{3x}\cos(2x)$: the annihilator is $\left[(D-3)^2 + 4\right]^2$, reflecting both the oscillation and the linear factor of x.

Practice Problems

One problem of each exam type from Unit 02. Work through these at the end.

P1. *Existence & Uniqueness — Interval of Solution*
Find the largest interval centered about $x = 0$ on which the IVP has a unique solution.

$$y'' + (\tan x)\, y = e^x, \qquad y(0) = 1, \quad y'(0) = 0.$$

P2. *Wronskian and Fundamental Set Verification*
Verify that $y_1 = x^2$ and $y_2 = x^4$ form a fundamental set of solutions of $x^2y'' - 5xy' + 8y = 0$ on $(0, \infty)$ by computing $W(x^2, x^4)$.

P3. *Reduction of Order*
Given that $y_1 = x^2$ solves $x^2y'' + 2xy' - 6y = 0$, use reduction of order to find a second solution $y_2(x)$.

P4. *Characteristic Equation — Repeated Root (IVP)*
Solve the initial value problem.

$$y''' + 6y'' + 9y' = 0, \qquad y(0) = 0, \quad y'(0) = 1, \quad y''(0) = -4.$$

P5. *Characteristic Equation — Higher Order*
Find the general solution of

$$\frac{d^4y}{dx^4} - 14\,\frac{d^2y}{dx^2} - 32y = 0.$$

P6. *Undetermined Coefficients — Superposition with Modification Rule (IVP)*
Solve the initial value problem.

$$y'' + 4y' + 4y = (7 + x)e^{-2x}, \qquad y(0) = 6, \quad y'(0) = 8.$$

P7. *Undetermined Coefficients — Resonance (IVP)*
Solve the initial value problem.

$$y'' + y = 10\cos(2x) - 4\sin(x), \qquad y\left(\tfrac{\pi}{2}\right) = -1, \quad y'\left(\tfrac{\pi}{2}\right) = 0.$$

P8. *Annihilator — Identify the Operator*
Find a linear differential operator that annihilates each function.

(a) $(7 - e^{x})^2$

(b) $8 + e^{x}\cos(5x)$

(c) $e^{-x}\sin(x) - e^{7x}\cos(x)$

P9. *Annihilator — Full Solution*
Use the annihilator method to find the general solution of

$$y'' + 3y' - 4y = 16e^{2x}.$$

P10. *Mixed Review — Third-Order Equation*
Find the general solution of

$$3y''' + 7y'' + 10y' + 8y = 0.$$

Hint: test $r = -4/3$.